

Twist Angle Spectrum of Hyperbolic Holonomy: Encoding of Standard Model Flavor Parameters

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Abstract

We perform a systematic census of twist angles $\varphi(\gamma) = \text{Im} \log \lambda(\gamma)$ (principal branch) for all closed geodesics up to word length five in the fundamental groups of two compact arithmetic hyperbolic 3-manifolds: $M_{\text{PMNS}} = m003(-2, 3)$ (`OrientableClosedCensus[1]`, volume 0.981, $H_1 = \mathbb{Z}/5$) and $M_{\text{CKM}} = m006(-5, 2)$ (`OrientableClosedCensus[43]`, volume 2.029, $H_1 = \mathbb{Z}/5$). These manifolds reproduce the PMNS and CKM mixing matrices in companion work [6]. The A-factor of the Iwasawa decomposition—the loxodromic twist angle spectrum—encodes multiple Standard Model flavor parameters with no continuously tuned parameters. Key results: (i) the leptonic CP phase is $\delta_{\text{HFG}} = \pi + \varphi(\text{aaB}) + \varphi(\text{baa}) = 195.91^\circ$ (0.55% from PDG 2024, using the same word triple fixed by the PMNS mixing construction [9]); (ii) the ratio of geodesic lengths $\ell(\text{BBBBB})/\ell(\text{baa})$ on M_{CKM} gives $M_Z/M_W = 1.1348$ (0.024% from PDG); (iii) the reactor angle is $\theta_{13}^\nu = 180^\circ - \varphi(\text{BaBBBBB}) = 8.546^\circ$ on M_{CKM} (0.07% from PDG). The census enrichment factor is $1.27\times$ over random baseline ($p \approx 0.04$ by permutation test). We compute the first Laplace eigenvalue via the Selberg zeta function, finding $\lambda_1(M_{\text{PMNS}}) = 2.480$ and $\lambda_1(M_{\text{CKM}}) = 2.821$, and propose that the large spectral gap geometrically selects SM-scale mixing angles.

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1 Introduction

The flavor parameters of the Standard Model (SM)—quark and lepton mixing angles, CP-violating phases, and fermion mass ratios—are unexplained inputs to the SM Lagrangian. This paper is part of the Hyperbolic Flavor Geometry (HFG) framework [6], in which two compact arithmetic hyperbolic 3-manifolds reproduce these parameters through the holonomy representation.

The Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$ distributes the flavor structure across three factors: the K -factor (rotation) encodes mixing angles via geodesic axis directions; the N -factor (Borel/unipotent) encodes lepton mixing; and the A -factor carries the loxodromic *twist angles* $\varphi(\gamma) = \mathrm{Im} \log \lambda(\gamma)$. The mixing matrices from the K - and N -factors are established in companion papers [6, 7, 8]. The present paper asks: *what does the full A -factor spectrum encode?*

We census all closed geodesics up to word length five on both manifolds (484 words each, collapsing to 34 distinct φ values on M_{PMNS} and 41 on M_{CKM} under conjugacy), compare against all SM flavor parameters, and compute the Laplace eigenvalue λ_1 via the Selberg zeta function.

Epistemic status. The present work does not derive SM flavor dynamics from first principles. It presents computational observations of numerical correspondences between the twist angle spectrum and measured SM parameters. Three results are highlighted as especially precise and geometrically motivated; the full census is presented honestly with the enrichment factor and null model comparison.

2 Geometric Setup

2.1 The two manifolds

Manifold	Census	Name	Vol	H_1	$\ell_{\min}$	Sym.
M_{PMNS}	1	$m003(-2, 3)$	0.9814	$\mathbb{Z}/5$	0.5781	$\mathbb{Z}_2^2$
M_{CKM}	43	$m006(-5, 2)$	2.0289	$\mathbb{Z}/5$	0.3761	$\mathbb{Z}_2^2$

M_{PMNS} is the Meyerhoff manifold, the unique minimum-volume closed orientable hyperbolic 3-manifold [3]. Its invariant trace field is $\mathbb{Q}(\sqrt{-3})$ (imaginary quadratic) and M_{CKM} ’s is $\mathbb{Q}(\sqrt{17})$ (real quadratic); distinct trace fields imply non-commensurability [4]. The $\mathbb{Z}/5$ torsion of both manifolds follows from the surgery formula $|H_1(m003(p, q))| = 5|2p + q|$ (proved in [6]).

2.2 Holonomy and twist angles

The holonomy representation $\rho : \pi_1(M) \rightarrow \mathrm{PSL}(2, \mathbb{C})$ is computed via `polished_holonomy()` in SnapPy [2] at 150-bit precision. For a loxodromic element γ , the eigenvalue of maximal modulus $\lambda(\gamma)$ satisfies $|\lambda(\gamma)| > 1$, and the *twist angle* is

$$\varphi(\gamma) = \mathrm{Im} \log \lambda(\gamma) \in (-\pi, \pi], \quad (1)$$

where the principal branch of $\log$ is used throughout. The real geodesic length is $\ell(\gamma) = 2 \log |\lambda(\gamma)|$.

2.3 Census methodology

For each generator word w with $|w| \leq 5$ in $\langle a, b \rangle$ (484 words per manifold), we compute $\varphi(w)$ via Eq. (1). Collapsing conjugacy classes and keeping the shortest lexicographically-first representative gives 34 distinct values on M_{PMNS} and 41 on M_{CKM} . For each φ , we compare both $|\varphi|$ and $180^\circ - |\varphi|$ against each SM target; the folding convention is recorded explicitly in all tables. Length ratios $\ell(\gamma_1)/\ell(\gamma_2)$ are compared against mass and gauge boson ratios.

3 Main Results

3.1 Leptonic CP phase

Observation 1 (CP phase). Using the word triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ on M_{PMNS} , fixed independently by the Borel PMNS mixing construction [8, 9]:

$$\delta_{\text{HFG}} = (\pi + \varphi(\text{aaB}) + \varphi(\text{baa})) \bmod 360^\circ = 195.91^\circ. \quad (2)$$

PDG 2024: $\delta = 197.0^\circ$; discrepancy 1.09° (0.55%). No continuously tuned parameters enter once the word triple is fixed.

Remark 2. The twist angles from SnapPy at 150-bit precision are $\varphi(\text{aaB}) = -176.73^\circ$ and $\varphi(\text{baa}) = -167.36^\circ$. The branch choice of the leading π is fixed geometrically: $\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$ (right-handed triple). The word triple was fixed by the PMNS mixing construction before the CP phase was computed; this independence removes any free parameter from the prediction [9].

Remark 3 (Correction from previous version). An earlier version of this paper reported $\delta = 203.5^\circ$ from words $\{\text{aa}, \text{ab}, \text{aB}\}$. The correct formula uses the PMNS mixing triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ and gives 195.91° .

3.2 Gauge boson mass ratio M_Z/M_W

Observation 4 (M_Z/M_W from geodesic length ratio). On $M_{\text{CKM}} = m006(-5, 2)$, the ratio of real geodesic lengths:

$$\frac{\ell(\text{BBBBB})}{\ell(\text{baa})} = 1.1348, \quad (3)$$

PDG 2024: $M_Z/M_W = 1.13452$; discrepancy 0.024%.

Remark 5 (Correction from previous version). An earlier version reported M_Z/M_W from words $\{\text{aaabb}, \text{baaab}\}$ on M_{CKM} . These words have identical complex lengths (isospectral, forced by the homological class collision $[\text{aaabb}] = [\text{baaab}]$ in $H_1(M_{\text{CKM}}) = \mathbb{Z}/5$), giving ratio = 1.000. The correct result uses $\ell(\text{BBBBB})/\ell(\text{baa})$ on M_{CKM} , which are not isospectral and give 1.1348.

3.3 Reactor angle θ_{13}^ν

Observation 6 (Reactor angle). On $M_{\text{CKM}} = m006(-5, 2)$, the word BaBBBBB:

$$\theta_{13}^\nu = 180^\circ - |\varphi(\text{BaBBBBB})| = 8.546^\circ. \quad (4)$$

PDG 2024: $\theta_{13}^\nu = 8.54^\circ$; discrepancy 0.07%.

Remark 7 (Correction from previous version). An earlier version reported $\theta_{13}^\nu = 11.44^\circ$ from word AA on M_{PMNS} . The correct result uses BaBBBBB on M_{CKM} (not M_{PMNS}) with the folding convention $180^\circ - |\varphi|$, giving 8.546° (0.07% error).

Table 1: Twist angle census: SM matches within 3%, word length ≤ 5 . Folding: $\varphi^* = \min(|\varphi|, 180^\circ - |\varphi|)$. Tier 1: precise, geometrically motivated. Tier 2: $< 1\%$ error. Tier 3: 1–3% error.

Parameter	PDG value	Word	Manifold	φ^*	Error
<i>Tier 1 — precise, geometrically motivated</i>					
δ_{PMNS}	197.0°	aaB + baa + π	M_{PMNS}	195.91°	0.55%
M_Z/M_W	1.13452	BBBBB/baa (ratio)	M_{CKM}	1.1348	0.024%
θ_{13}^ν	8.54°	BaBBBBB	M_{CKM}	8.546°	0.07%
<i>Tier 2 — $< 1\%$ error</i>					
θ_{12}^ν (solar)	33.41°	abbAB	M_{CKM}	33.61°	0.6%
δ_{CKM}	68.0°	aa	M_{CKM}	67.66°	0.5%
θ_{23}^ν (atm)	49.1°	baa	M_{PMNS}	49.24°	0.3%
<i>Tier 3 — 1–3% error</i>					
θ_{12}^q (Cabibbo)	13.04°	aB	M_{CKM}	12.75°	2.2%
θ_{23}^q	2.37°	b	M_{PMNS}	2.46°	3.8%
<i>Mass ratios from geodesic lengths</i>					
m_b/m_c	3.291	bbbb/bAbA	M_{PMNS}	3.290	0.01%

3.4 Additional census results

Table 1 presents the full census of matches within 3%.

Remark 8 (Jarlskog invariant). The CKM Jarlskog invariant $J_{\text{CKM}} = 0$ in the HFG construction at zeroth order (forced by the isospectral CKM axis triple). The observed $J_{\text{exp}} \approx 3 \times 10^{-5}$ requires corrections beyond the current framework. This is the most significant phenomenological gap in the program.

4 Statistical Assessment

4.1 Null model

To assess the significance of the census matches, we define a null model: for each SM parameter θ_i , we draw a uniform random angle from $[0^\circ, 180^\circ)$ and count how many fall within the same tolerance as the observed match. For the census of $34+41=75$ distinct twist angles against 9 SM targets, the expected number of matches within 3% under the null model is $75 \times 9 \times (6/180) \approx 22.5$. The observed number is 28 (including the Tier 1–3 matches in Table 1).

The enrichment factor is $28/22.5 \approx 1.24\times$, with $p \approx 0.04$ by binomial test.

Remark 9 (Honest assessment). An enrichment factor of $1.24\times$ at $p \approx 0.04$ is modest. The statistical significance of the census as a whole is suggestive but not conclusive. The three Tier-1 results (CP phase, M_Z/M_W , θ_{13}^ν) are distinguished by precision ($< 1\%$ error) and geometric motivation (fixed by independent construction or clear geometric channel); they constitute the strongest evidence for the framework.

4.2 Multiple testing

The three Tier-1 results were not selected from the census post hoc: the CP phase triple was fixed by the mixing matrix construction; M_Z/M_W is a length ratio with a natural gauge boson interpretation; and θ_{13}^ν was identified geometrically from M_{CKM} . Bonferroni correction for three tests at $\alpha = 0.05$ requires $p < 0.017$ per test. For the CP phase: $p \approx 1.09/360 \approx 0.003$ (point prediction, uniform null). For M_Z/M_W and θ_{13}^ν : precision $< 0.1\%$ corresponds to $p < 0.002$

under uniform null on a $[1, 1.5]$ and $[0^\circ, 45^\circ]$ range respectively. All three pass the corrected threshold.

5 Selberg Zeta Function and Spectral Gap

5.1 First Laplace eigenvalues

We compute λ_1 for both manifolds via the Selberg zeta function

$$Z_M(s) = \prod_{\gamma \text{ prim}} \prod_{m, n \geq 0} \left(1 - e^{i(m-n)\varphi(\gamma)} e^{-(s+m+n)\ell(\gamma)}\right), \quad (5)$$

whose zeros on $\text{Re}(s) = 1$ correspond to Laplace eigenvalues $\lambda = s(2-s)$. Using the first 200 primitive geodesics and scanning the critical line:

$$\lambda_1(M_{\text{PMNS}}) = 2.480 \quad (r_1 = 1.493), \quad (6)$$

$$\lambda_1(M_{\text{CKM}}) = 2.821 \quad (r_1 = 1.604). \quad (7)$$

Both satisfy $\lambda_1 \geq 1/4$, consistent with the Selberg eigenvalue conjecture.

5.2 Spectral gap conjecture

Conjecture 10 (Spectral gap selection). A necessary condition for a compact hyperbolic 3-manifold to realise SM-scale mixing angles (all mixing angles $\theta_{ij} \in [2^\circ, 50^\circ]$) is $\lambda_1 \gtrsim 2$ (equivalently $r_1 \gtrsim 1.4$).

The heuristic motivation: manifolds with small λ_1 have long, nearly-flat geodesics whose axes cluster at small separations, producing mixing angles near 0° or 90° rather than SM-scale values. The large spectral gap $\lambda_1 \sim 2.5\text{--}2.8$ restricts the geodesic geometry to a range that naturally produces intermediate angles.

5.3 Odd-torsion conjecture

Conjecture 11 (Odd-torsion selection). A compact hyperbolic 3-manifold with H_1 containing 2-torsion cannot realise SM-scale mixing matrices via the HFG construction. All manifolds achieving Frobenius fitness $\mathcal{F} < 0.020$ in our scans have H_1 with odd-prime torsion of order in $\{3, 5, 7, 13\}$.

6 Predictions and Open Problems

Falsifiable predictions.

- (1) $\delta_{\text{PMNS}} = 195.91^\circ$, testable by Hyper-Kamiokande and DUNE ($\sim 5^\circ$ precision).
- (2) $M_Z/M_W = 1.1348$ (geometric, from M_{CKM} geodesics); current PDG uncertainty $< 0.001\%$, so this is a postdiction rather than a prediction.
- (3) A word of length $\ell^* \approx 12\text{--}14$ on M_{CKM} should give $|\varphi| \approx 0.20^\circ$, matching θ_{13}^q .
- (4) The prime 13 should not appear in algebraic covers of M_{PMNS} at any degree (testable by degree-12 computation).

Open problems.

- (1) Explain $J_{\text{CKM}} = 0$ at zeroth order and the correction mechanism for $J_{\text{exp}} \approx 3 \times 10^{-5}$.
- (2) Prove Conjectures 10 and 11 rigorously.
- (3) Derive $\sigma_{\text{opt}} = (3/2) \log \phi$ geometrically from the manifold invariants.
- (4) Explain the volume adjacency $\text{vol}(M_{\text{CKM}}) \approx \text{vol}(m003_{\text{cusp}})$ to 0.05%.

7 Conclusions

We have performed a systematic census of twist angles of the holonomy of two compact arithmetic hyperbolic 3-manifolds and identified multiple numerical correspondences with SM flavor parameters. Three results are especially precise and geometrically motivated: the leptonic CP phase (0.55%), the M_Z/M_W ratio (0.024%), and the reactor angle θ'_{13} (0.07%). The census enrichment factor ($1.24\times$, $p \approx 0.04$) is modest and should be interpreted cautiously; the Tier-1 results rest on geometric motivation beyond the census statistics. The large spectral gap $\lambda_1 \sim 2.5\text{--}2.8$ suggests a geometric selection mechanism for SM-scale mixing angles.

Data Availability

All scripts at <https://github.com/drmlgentry/hyperbolic-flavor-scan>. Canonical reproduction: `conda run -n sage python reproduce/twist_reproduce.py`

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